TECHNICAL SPECIFICATION & RESEARCH BLUEPRINT

Machine Learning Architecture & Engineering Specification

Deep-dive into KnowledgeTrace MCQ: Cognitive Diagnosis Models, Bayesian Knowledge Tracing, Topological DAGs, Telemetry Flows, Mathematical Formulations, and the State-of-the-Art ML Roadmap.

Target System: KnowledgeTrace MCQ	**Core Models:** BKT + Bayesian DINA	**Graph Validation:** NetworkX Poset DAG
Inference Latency: < 15ms (Vectorized)	**Confidence Metric:** Quadratic Entropy	**Roadmap:** pyBKT EM + 2PL IRT + AKT

1. High-Level ML Architecture Design

KnowledgeTrace MCQ decouples the educational evaluation into a **3-Tier Hybrid AI & Psychometrics Architecture** that provides interpretable, fine-grained concept mastery while maintaining mathematical guarantees against circular curricula:

- **Layer 1 — Curriculum Topology (Structural Learning):** LLM-based prerequisite edge inference using Groq API (groq/compound), validated by NetworkX to guarantee a strict Directed Acyclic Graph (DAG) with zero circular dependencies.
- **Layer 2 — Sequential Mastery Dynamics (Temporal Learning):** A 2-state Hidden Markov Model implementing Bayesian Knowledge Tracing (BKT) to update concept mastery $P(L_t)$ dynamically after every response sequence.
- **Layer 3 — Multi-Attribute Cognitive Diagnosis (Structural Diagnosis):** A discrete multi-skill Cognitive Diagnosis Model (DINA) using Q-matrix mapping and connected-block graph partitioning for exact Bayesian posterior marginalization.

2. End-to-End Reactive Telemetry & Data Flow

Every student submission initiates an automated, closed-loop pipeline across PostgreSQL, NumPy, and the UI:

- **Step 1 — Q-Matrix Definition:** Teachers construct tests and associate questions with one or more concepts in question_skills ($q_{jk} \in \{0, 1\}$).
- **Step 2 — Telemetry Ingestion:** When a student answers a question, the client captures correctness (X_j), option selected, and reaction time in milliseconds (response_time_ms), saving to responses.
- **Step 3 — BKT Forward Trigger:** update_mastery() runs the BKT forward algorithm over all chronological attempts for $(student, concept)$, persisting the updated $P(L_t)$ to student_concept_mastery.
- **Step 4 — DINA Graph Decomposition:** update_diagnosis() creates a co-occurrence graph of skills, performs BFS connected-component partitioning, computes exact joint posteriors, and writes to cognitive_diagnosis_profile.
- **Step 5 — Diagnostic Visualization:** The frontend renders mastery probability, confidence ratings (LOW, MEDIUM, HIGH), and flags root-cause missing prerequisites.

3. Deep-Dive: The Four Core ML Models

Model 1: LLM Prerequisite Graph Generator + NetworkX Invariant

Concepts are not isolated; they form prerequisite chains ($A \to B$). Circular dependencies ($A \to B \to C \to A$) break topological sequencing. The system prompts Groq LLM to infer candidate edges, then provisionally adds them to an in-memory networkx.DiGraph. If nx.is_directed_acyclic_graph(G) evaluates to False, the edge is rejected.

Model 2: Bayesian Knowledge Tracing (BKT)

BKT models the latent knowledge state $L_t \in \{0, 1\}$ as an HMM. Parameters are calibrated for 4-option MCQs:

Parameter	Symbol	Value	Psychometric Meaning
Prior Knowledge	**P(L0)**	0.20	Probability of knowing the concept before the first attempt.

Transition Rate	**P(T)**	0.15	Probability of learning the concept between practice opportunities.
Slip Rate	**P(S)**	0.10	Probability of making a careless mistake despite knowing the concept.
Guess Rate	**P(G)**	0.25	Probability of answering correctly by chance on a 4-option MCQ.

Model 3: Multi-Attribute DINA Cognitive Diagnosis Model

DINA operates on a **conjunctive assumption**: a question requiring multiple concepts can only be mastered if the student masters *all* required concepts. To eliminate the exponential complexity of 2^K latent profiles for large curriculums, the engine partitions the Q-matrix into connected skill blocks using BFS. Exact vectorized NumPy marginalization is performed for $K \le 12$, with an exact single-skill fallback for $K > 12$.

Model 4: Epistemic Confidence & Reliability Tiers

Confidence is computed via quadratic entropy: **Confidence = 1 - 4·p·(1 - p)**. Maximum uncertainty occurs at $p = 0.50$ (Score = 0.0), while definitive mastery ($p=1.0$) or definitive deficit ($p=0.0$) yields a Confidence of 1.0. Tiers: **LOW** (< 0.40), **MEDIUM** (0.40–0.75), and **HIGH** (≥ 0.75).

4. Mathematical Calculations & Formulas

BKT Observation Update (Bayes' Rule) & Latent Transition:

```
If Correct (X_t = 1):
  P(L_t | X_t=1) = [ P(L_t) * (1 - P(S)) ] / [ P(L_t)*(1 - P(S)) + (1 - P(L_t))*P(G) ]

If Incorrect (X_t = 0):
  P(L_t | X_t=0) = [ P(L_t) * P(S) ]       / [ P(L_t)*P(S)       + (1 - P(L_t))*(1 - P(G)) ]

Transition Step:
  P(L_t+1)       = P(L_t | X_t) + [ 1 - P(L_t | X_t) ] * P(T)
```

Empirical BKT Verification (4 Attempts: [Correct, Incorrect, Correct, Correct]):

Step	Prior P(L)	Observation	Posterior P(L|X)	Updated P(L_t+1)
Initial	0.2000	—	—	**0.2000**
t = 1	0.2000	Correct (1)	0.4737	**0.5526**
t = 2	0.5526	Incorrect (0)	0.1418	**0.2705**
t = 3	0.2705	Correct (1)	0.5714	**0.6357**
t = 4	0.6357	Correct (1)	0.8653	**0.8855**

DINA Conjunctive Latent Response & Log-Sum-Exp Marginalization:

```
Ideal Latent Response:
  ξ_j(α) = ∏ [ α_k ]^(q_jk)   => 1 if student has ALL required skills, 0 otherwise.

Item Likelihood:
  P(X_j = 1 | α) = (1 - s_j)^(ξ_j(α)) * (g_j)^(1 - ξ_j(α))

Joint Posterior Marginalization:
  log P(α | X)   = log P(α) + Σ log P(X_j | α)  - LogSumExp[ log P(α' | X) ]
  P(α_k = 1 | X) = Σ_{α : α_k = 1} P(α | X)
```

5. How Models are Built vs. Trained

Currently, both BKT and DINA leverage **cold-start informative priors** ($P(L_0)=0.20, P(T)=0.15, s=0.10, g=0.25$). The models do not require offline training to deliver immediate value. At runtime, inference is purely online and analytical, updating probabilities in under 15ms per submission without expensive gradient updates.

6. Performance Characteristics: Strengths vs. Limitations

Dimension	Current System Strengths	Current Architectural Limitations
Explainability	100% white-box probabilities; teachers can trace why a score changed.	Discrete binary skills; cannot model nuanced partial credit.
Cold-Start	Zero training delay; works immediately on any curriculum.	Identical slip/guess parameters across easy and hard questions.
Telemetry	Captures millisecond reaction times accurately.	Reaction time is stored in DB but not yet fused into probability math.
Scale	Graph block partitioning avoids 2^K state explosions for $K \le 12$.	Falls back to single-skill independence if block exceeds 12 concepts.

7. Concrete ML Improvements & Research Roadmap

• **1. Concept-Specific Expectation-Maximization (EM):** Use the Baum-Welch algorithm via the pyBKT library to learn distinct parameters $(P(L_0), P(T), P(S), P(G))$ per concept from historical responses.

• **2. Chronometric Response-Time Infusion:** Modulate guessing probability g_j using log-normal response time modeling: answers given in < 3 seconds elevate the guess rate, penalizing blind lucky guesses.

• **3. 2-Parameter Logistic (2PL) Item Response Theory:** Estimate continuous latent student ability θ_i along with question difficulty b_j and discrimination a_j: $P(X=1|\theta) = [1 + e^{-a(b - \theta)}]^{-1}$.

• **4. Deep Knowledge Tracing (DKT / AKT):** Implement Self-Attentive Knowledge Tracing (AKT) to capture long-range contextual decay across extensive question sequences.

• **5. Graph Neural Networks (GNN / R-GCN) over the DAG:** Propagate mastery backward and forward across the concept DAG: mastering a high-level skill reinforces prerequisites, while failing prerequisites caps target skills.

8. Key Codebase Files & Implementation References

• **backend/app/services/mastery_service.py**: Abstract base estimator & BKT forward sequence algorithm.
• **backend/app/services/dina_service.py**: Multi-attribute DINA cognitive diagnosis & graph-block marginalization.
• **backend/app/services/concept_graph.py**: Groq LLM prerequisite extraction & NetworkX DAG invariant verification.
• **backend/app/models.py**: Schemas for StudentConceptMastery, QuestionSkill, and CognitiveDiagnosisProfile.